Boxhunter erection site lifting equipment **ANNEX 2**

Reference: Boxhunter site erection drawings G1719-SE2

<u>Summary of lifting equipment requirements</u>

1 pc	Manlift	duration about 30 days
1 pc	Forklift truck 4t capacity	duration about 14 days
1 pc	Forklift truck 16t capacity	duration about 14 days
1 pc	Mobile crane 150-200 t capacity	duration about 7 days

<u>Typical lifting equipment:</u>

Mobile cranes: 150tn -200tn

Mobile cranes are needed at times as per site schedule/plan with the following lifting capacities (Note! The following information indicates guidelines only for the mobile crane selection and will give some idea about the mobile crane needs. The crane capacities indicated below depend somewhat also on the mobile crane type and the lower figure given may NOT be quite enough in most cases!):

- One 150 Ton capacity mobile crane for one day RTG crane erection as per schedule (this size of mobile crane will reach about 8 metric ton trolley up to height of about 27 meters for 1 over 5 Boxhunter crane). Typically crane erection should be done on one day but it is good to reserve additional day if wind conditions prevents to finalize crane erection. 8m/s wind speed is max. wind speed when crane erection is allowed

- Also this 150tn mobile crane can be used for installing sill beams (access side & machinery side) to leg frames during erection with the help of lifting frame see below picture

The mobile crane size given above will give some idea about capacities required for Boxhunter crane erection. The decision on what size of mobile crane(s) will be used for unloading of cargo, for lifting the trolley onto preassembly stand and for the final erection will depend on number of things which will have to be studied case by case. The heaviest item is the trolley and spreader (approx. 9 metric tons) and will therefore be decisive for the mobile crane capacity requirement and outreach/ crane position. The following may be considered when selecting lifting equipment: what mobile cranes are available near site location, outreach radius and height at required capacity, transport costs (one bigger crane or two smaller) and possibilities to use small radius for heavy lifts on the job site.

Forklifts:

Forklift(s) are needed from the very start, first for establishing the job site and positioning tool cabins etc., receiving/unloading deliveries, laying out crane parts and then in lot of occasions during the entire erection. It is advisable to have one Forklift/Telescope Forklift of 4 Ton capacity for the entire duration of the erection as per site schedule. There may be need for a second (bigger) forklift for a short time during unloading the delivery to handle main girders, leg sections, diesel alternator (6 tons), machinery side sill beam (6 tons) and spreader (8-9 tons). Later on there is a need for this bigger size forklift when installing diesel alternator (6 tons) and sill beams (5 tons) right after the erection of steel structures (schedule item Erection of crane).

Man lift:

Man lift is needed continuously starting from the crane erection phase, i.e. for seven to eight days per Boxhunter crane. Lifting height of the man lift have to be 27 meters. This is essential e.g. for main girder flange joints bolt tightening. Man basket does not provide enough reach for main girder inner side flange bolt torqueing.


KONECRANES